

Lab Assignment -1

Q1. Develop the computer program for the overall design of transformer. Choose transformer of appropriate kVA rating. Prepare also the design sheet in A3 paper.

Core & Window Design Inputs:

- **KVA rating:** 25.00
- **Line frequency (Hz):** 50
- **Number of phases:** 3
- **Voltage per turn (Et):** 2.25
- **Flux density (Tesla):** 1.0000
- **Stacking factor:** 0.90
- **Type of core:** 2 (2-Stepped)
- **Height of window (mm):** 300
- **Width of window (mm):** 120
- **Window space factor (Kw):** 0.180
- **Ratio - area of yoke to limbs:** 1.2
- **Depth of yoke (mm):** 114

Winding Design Inputs (LV & HV)

- **Secondary line voltage (V):** 433.0
- **Secondary connection type:** 1 (Star)
- **Secondary current density (A/mm²):** 2.23
- **Secondary conductor area (mm²):** 14.493
- **Radial depth of LV winding (mm):** 9.1000
- **Axial height of LV winding (mm):** 285.00
- **Mean length of turn for LV (mm):** 0.462 (Note: Input as mm)
- **Primary line voltage (V):** 11000
- **Primary connection type:** 2 (Delta)
- **Primary current density (A/mm²):** 2.43

- **Primary conductor area (mm²):** 0.316
- **Radial depth of HV winding (mm):** 26.220
- **Axial height of HV winding (mm):** 220.320
- **Mean length of turn for HV (mm):** 0.6667

Electrical & Loss Inputs

- **Width of insulation duct (mm):** 5
- **Resistivity of HV conductor:** 0.021
- **Resistivity of LV conductor:** 0.021
- **Total weight of limbs (kg):** 69.324
- **Specific iron loss for limbs (W/kg):** 1.2
- **Total weight of yoke (kg):** 115.182
- **Specific iron loss for yoke (W/kg):** 0.850

Program Code:

```
import math
print('TRANSFORMER DESIGN CALCULATIONS (A.K. SAWHNEY - Using Python)')
print('='*50)

# CORE DESIGN
print('CORE DESIGN')
kva = float(input('KVA rating: '))
f = float(input('Line frequency (Hz): '))
m = float(input('Number of phases: '))

# Using Voltage per Turn directly as per design sheet
et = float(input('Voltage per turn (Et): '))

bm = float(input('Flux density (Tesla): '))
ki = float(input('Stacking factor: '))

phlm = et / (4.44 * f)
ai_m2 = phlm / bm
ai_mm2 = ai_m2 * 1e6

print('Enter type of core: 1)Square 2)2-Stepped 3)3-Stepped 4)4-Stepped')
c = int(input('Choice: '))
ct_map = {1: 0.45, 2: 0.56, 3: 0.6, 4: 0.62}
```

```
ct = ct_map.get(c, 0.45)
d_mm = math.sqrt(ai_mm2 / ct)
print(f'Diameter of circumscribing circle: {d_mm:.2f} mm')
```

WINDOW DESIGN

```
print('\nWINDOW DESIGN')
hw_mm = float(input('Height of window (mm): '))
ww_mm = float(input('Width of window (mm): '))
kw_factor = float(input('Window space factor (Kw): '))
```

YOKE DESIGN

```
print('\nYOKE DESIGN')
ratioyl = float(input('Ratio - area of yoke to limbs: '))
dy_mm = float(input('Depth of yoke (mm): '))
ay_mm2 = ratioyl * ai_mm2
hy_mm = (ay_mm2 / ki) / dy_mm
print(f'Yoke height: {hy_mm:.4f} mm')
```

LOW VOLTAGE (LV) WINDING DESIGN

```
print('\nLOW VOLTAGE (LV) WINDING DESIGN')
v_line_s = float(input('Secondary line voltage (V): '))
c1 = int(input('Type of connection: 1.Star 2.Delta: '))
vsp = v_line_s / math.sqrt(3) if c1 == 1 else v_line_s
ts = round(vsp / et)
isp = (kva * 1000) / (m * vsp)
```

```
delta_s = float(input('Secondary current density (A/mm^2): '))
as_area = float(input('Secondary conductor area (mm^2): '))
bs_mm = float(input('Radial depth of LV winding (mm): '))
h_lv_mm = float(input('Axial height of LV winding (mm): '))
mlt_s_mm = float(input('Mean length of turn for LV (mm): '))
```

HIGH VOLTAGE (HV) WINDING DESIGN

```
print('\nHIGH VOLTAGE (HV) WINDING DESIGN')
v_line_p = float(input('Primary line voltage (V): '))
c2 = int(input('Type of connection: 1.Star 2.Delta: '))
vpp = v_line_p / math.sqrt(3) if c2 == 1 else v_line_p
tp = round(ts * (vpp / vsp))
ipp = (kva * 1000) / (m * vpp)
```

```
delta_p = float(input('Primary current density (A/mm^2): '))
ap_area = float(input('Primary conductor area (mm^2): '))
bp_mm = float(input('Radial depth of HV winding (mm): '))
h_hv_mm = float(input('Axial height of HV winding (mm): '))
mlt_p_mm = float(input('Mean length of turn for HV (mm): '))
```

INSULATION & DUCT

```

print('\nINSULATION')
cly_mm = float(input('Width of insulation duct between LV and HV (mm): '))

# ELECTRICAL PARAMETERS
print('\nELECTRICAL PARAMETERS')
rop = float(input('Resistivity of HV conductor (ohm-mm^2/m): '))
ros = float(input('Resistivity of LV conductor (ohm-mm^2/m): '))

# Resistance referred to primary
rp = (tp * (mlt_p_mm / 1000) * rop) / ap_area
rs = (ts * (mlt_s_mm / 1000) * ros) / as_area
ref = rp + (rs * (tp / ts)**2)
ep = (ipp * ref) / vpp

# Leakage Reactance (Sawhney Formula)
lmt_avg_m = (mlt_p_mm + mlt_s_mm) / (2 * 1000)
lc_avg_m = (h_lv_mm + h_hv_mm) / (2 * 1000)
mu_0 = 4 * math.pi * 1e-7

# Duct term in meters: Width + (Depth_LV + Depth_HV) / 3
duct_term_m = (cly_mm + (bs_mm + bp_mm) / 3) / 1000
xp = (2 * math.pi * f * mu_0 * (tp**2) * lmt_avg_m * duct_term_m) / lc_avg_m
epx = (ipp * xp) / vpp
epi = math.sqrt(ep**2 + epx**2)

# LOSSES AND EFFICIENCY
print('\nLOSSES AND EFFICIENCY')
w_limbs = float(input('Total weight of limbs (kg): '))
sl_limbs = float(input('Specific iron loss for limbs (W/kg): '))
w_yoke = float(input('Total weight of yoke (kg): '))
sl_yoke = float(input('Specific iron loss for yoke (W/kg): '))
core_loss = (w_limbs * sl_limbs) + (w_yoke * sl_yoke)
cu_loss = m * (ipp**2 * ref)
total_loss = core_loss + cu_loss*1000
efficiency = (kva * 1000 / (kva * 1000 + total_loss)) * 100
print('\n' + '='*40)
print('FINAL CALCULATED PARAMETERS')
print('='*40)
print(f'P.U. Resistance (ep): {ep:.5f}')
print(f'P.U. Reactance (epx): {epx:.5f}')
print(f'P.U. Impedance (epi): {epi:.5f}')
print('='*40)
print(f'Total Core Loss: {core_loss:.2f} W')
print(f'Total Copper Loss: {cu_loss:.2f} KW')
print(f'Total Loss: {total_loss:.2f} W')
print(f'Full Load Efficiency: {efficiency:.2f} %')
print('='*40)

```

```

...  TRANSFORMER DESIGN CALCULATIONS (A.K. SAWHNEY - Using Python)
=====
CORE DESIGN
KVA rating: 25.00
Line frequency (Hz): 50
Number of phases: 3
Voltage per turn (Et): 2.25
Flux density (Tesla): 1.0000
Stacking factor: 0.90
Enter type of core: 1)Square 2)2-Stepped 3)3-Stepped 4)4-Stepped
Choice: 2
Diameter of circumscribing circle: 134.53 mm

WINDOW DESIGN
Height of window (mm): 300
Width of window (mm): 120
Window space factor (Kw): 0.180

YOKE DESIGN
Ratio - area of yoke to limbs: 1.2
Depth of yoke (mm): 114
Yoke height: 118.5396 mm

LOW VOLTAGE (LV) WINDING DESIGN
Secondary line voltage (V): 433.0
Type of connection: 1.Star 2.Delta: 1
Secondary current density (A/mm^2): 2.23
Secondary conductor area (mm^2): 14.493
Radial depth of LV winding (mm): 9.1000
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HIGH VOLTAGE (HV) WINDING DESIGN
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Type of connection: 1.Star 2.Delta: 2
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INSULATION
Width of insulation duct between LV and HV (mm): 5

ELECTRICAL PARAMETERS
Resistivity of HV conductor (ohm-mm^2/m): 0.021
Resistivity of LV conductor (ohm-mm^2/m): 0.021

LOSSES AND EFFICIENCY
Total weight of limbs (kg): 69.324
Specific iron loss for limbs (W/kg): 1.2
Total weight of yoke (kg): 115.182
Specific iron loss for yoke (W/kg): 0.850

=====
FINAL CALCULATED PARAMETERS
=====
P.U. Resistance (ep): 0.00002
P.U. Reactance (epx): 0.00002
P.U. Impedance (epi): 0.00003
-----
Total Core Loss:      181.09 W
Total Copper Loss:    0.62 KW
Total Loss:           801.35 W
Full Load Efficiency: 96.89 %
=====

```

Output

Measurements:

- Diameter of circumscribing circle: 134.53 mm
- Yoke height (calc.): 118.5396 mm
- P.U. Resistance: 0.00002
- P.U. Reactance: 0.00002
- P.U. Impedance: 0.00003
- Total Core Loss: 181.09W
- Total Copper Loss: 0.62KW
- Total Loss: 801.35W
- Full Load Efficiency: 96.89%